

ABHIMUNNYA DEY

Software Engineer — Full Stack Java Developer

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SUMMARY

Software Engineer with hands-on experience in full-stack development using Java, JavaScript, and modern frameworks including Spring Boot and React. Skilled in Data Structures & Algorithms, Object-Oriented Programming, and REST API design, with a track record of building scalable backend systems and responsive web interfaces. Currently gaining professional experience as a Software Engineer Intern, contributing to full-stack web application development.

SKILLS

Languages: Java · JavaScript

Frameworks: Spring Boot · Spring Data JPA · Spring Security · JWT · React

Databases: PostgreSQL

Concepts: Data Structures & Algorithms · OOP · Collections Framework · Full-Stack Development · REST API Design

Tools: Git · Postman

PROJECTS

Offline UPI Payment Mesh – Distributed Encrypted Transaction Relay System

GitHub

Distributed backend system that settles encrypted offline UPI payments through a Bluetooth mesh relay network with exactly-once transaction guarantees.

- Backend system that enables UPI payments without internet by relaying encrypted payment packets across a Bluetooth mesh network until a bridge node settles the transaction against a PostgreSQL ledger.
- Hybrid RSA-OAEP + AES-GCM encryption, atomic idempotency via ConcurrentHashMap putIfAbsent, optimistic locking via @Version, gossip protocol with TTL-based propagation, and a 5-stage ingestion pipeline covering deduplication, decryption, freshness validation, and atomic settlement.
- Eliminates 100% of duplicate settlements, replay attacks, and balance corruption under concurrent multi-node delivery – solving complete payment failure in zero-connectivity environments.

Learnova – AI-Powered E-Learning Platform

GitHub

Full-stack role-based e-learning platform

- Full-stack e-learning platform with role-based access control (STUDENT/ADMIN), supporting course management, lesson tracking, enrollment, quiz evaluation, and peer reviews. Structured as a RESTful microservice-ready backend with a React-based SPA frontend.
- Implemented stateless JWT authentication using HMAC-SHA256 signing algorithm, role-based authorization via Spring Security filter chain, ORM-based relational data persistence, and Redis cache-aside pattern to reduce redundant database queries.
- Addresses the lack of a centralized platform for structured course delivery, student progress tracking, and quiz-based assessment in digital education.

EXPERIENCE

Omnivera Technologies Pvt. Ltd

Software Engineer Intern

Remote

07/2026 – Present

- Developed responsive and tablet-optimized web interfaces using React.js, JavaScript, and Tailwind CSS, applying reusable component-based architecture, responsive design, and maintainable coding practices.
- Currently working on backend and database development using Node.js, Express.js, and MongoDB, with exposure to RESTful API design, authentication/authorization, data modeling, and frontend-backend integration.

EDUCATION

Bengal Institute Of Technology

B.Tech in Computer Science · CGPA: 7.15

2023 – 2027

LANGUAGES

Bengali: Native **English:** Proficient **Hindi:** Proficient **German:** Beginner